青蒿一握，济世千万

A Handful of Wormwood That Saved Millions



古方里没有煮沸这一步，怎样启发了青蒿素研究？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Project 523, a covert military mission involving many institutes, was named after its launch date, 23 May 1967. Its urgent target was the chloroquine-resistant malaria weakening troops in the Vietnam War. In January 1969 Tu Youyou was entrusted with leading her institute's research group, despite having no doctorate or foreign training.

In an exhaustive search, her team combed over 2,000 folk recipes, compiled 640 prescriptions, and evaluated about 380 extracts from some 200 medicinal herbs. Qinghao, or sweet wormwood, showed flashes of promise, yet its results were erratic. Seeking a clue, Tu revisited the classical texts.

Around AD 340, the physician Ge Hong's handbook advised taking "a handful of qinghao immersed in two litres of water". This bypassed boiling. Tu suspected heat during extraction was degrading a delicate component, so she switched to ether, a volatile solvent boiling at about 35 degrees Celsius. On 4 October 1971 sample 191 completely inhibited the parasite in infected mice.

Efficacy in mice indicated nothing about human safety. In July 1972, to assess its toxicity, Tu and two colleagues voluntarily took the extract. Thereafter, from August to October, it was administered to 21 malaria patients in Hainan, and on 8 November her team obtained pure crystals of the substance later named artemisinin.

In September 2011 Tu was named a Lasker Award recipient; its jury credited the drug with saving millions. On 5 October 2015 she won half the Nobel Prize in Physiology or Medicine, the first Chinese woman Nobel laureate. Since 2006 WHO guidelines have endorsed artemisinin-based combination therapies against malaria's deadliest form. The essence of Ge Hong's line lay in the boiling it skipped.

中文译文

“523项目”是一项多家研究机构参与的秘密军事任务，名称来自启动日期：1967年5月23日。项目急需解决的是对氯喹产生耐药性的疟疾，当时这种疾病正削弱越南战争中的部队。1969年1月，尽管没有博士学位，也未在国外受过训练，屠呦呦仍受命领导所在研究所的研究组。

团队进行了全面搜寻，仔细翻检2000多份民间药方，汇编640个方剂，评估了来自约200种药用植物的约380种提取物。青蒿，也就是黄花蒿，时而显出效果，结果却不稳定。为寻找线索，屠呦呦重新翻阅古代文献。

公元340年前后，医师葛洪的手册建议取“一把浸在两升水中的青蒿”。这种做法不经煮沸。屠呦呦怀疑，提取过程中的高温会破坏某种不耐热的成分，于是改用乙醚：一种沸点约35摄氏度、容易挥发的溶剂。1971年10月4日，第191号样品完全抑制了感染小鼠体内的疟原虫。

小鼠身上有效，并不能说明对人体是否安全。1972年7月，为评估毒性，屠呦呦与两名同事自愿服下提取物。随后，8月至10月间，提取物用于海南21名疟疾患者；11月8日，团队取得了这种物质的纯净晶体，它后来被命名为青蒿素。

2011年9月，屠呦呦获拉斯克奖，评审团认为这种药物挽救了数百万人的生命。2015年10月5日，她获得当年诺贝尔生理学或医学奖的一半份额，成为首位中国女性诺贝尔奖得主。自2006年起，世卫组织指南就推荐以青蒿素为基础的多药联合疗法，对抗最致命的一类疟疾。葛洪那句话的关键，恰在于没有煮沸。

核心词汇

先记这22项。连着下方的原文短语一起回想。

resistant adj.
有抗药性的
chloroquine-resistant malaria

be entrusted with 动词短语
受托承担；受命负责
entrusted with leading

exhaustive adj.
全面彻底的
In an exhaustive search

comb v.
仔细搜寻；翻检
combed over 2,000 folk recipes

compile v.
汇编
compiled 640 prescriptions

evaluate v.
评估
evaluated about 380 extracts

extract n.
提取物
evaluated about 380 extracts

erratic adj.
不稳定的；时好时坏的
its results were erratic

extraction n.
提取
heat during extraction

degrade v.
降解；使成分受破坏
was degrading a delicate component

component n.
成分
a delicate component

volatile adj.
易挥发的
a volatile solvent

solvent n.
溶剂
a volatile solvent

inhibit v.
抑制
completely inhibited the parasite

efficacy n.
药效；效力
Efficacy in mice

assess v.
评估
to assess its toxicity

toxicity n.
毒性
to assess its toxicity

administer … to … 动词短语
给……用药
administered to 21 malaria patients

substance n.
物质
pure crystals of the substance

endorse v.
认可；推荐
have endorsed artemisinin-based

combination n.
联合；合用
combination therapies

essence n.
关键；精髓
The essence of Ge Hong's line

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

covert adj.
秘密的；隐蔽的

military adj.
军事的

mission n.
任务

urgent adj.
紧迫的

malaria n.
疟疾

weaken v.
削弱

troop n.
部队

Vietnam War 专名
越南战争

despite prep.
尽管

doctorate n.
博士学位

folk adj.
民间的

prescription n.
药方

medicinal adj.
药用的

herb n.
草药；药用植物

sweet wormwood 名词短语
黄花蒿

flash n.
短暂显现

promise n.
有效的希望；前景

clue n.
线索

revisit v.
重新审阅

classical adj.
古代经典的

handbook n.
手册

be immersed in 动词短语
浸在……中

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

litre n.
升

bypass v.
避开；略过

delicate adj.
易受损的；此处指不耐热的

switch to 动词短语
改用

ether n.
乙醚

degree Celsius 名词短语
摄氏度

parasite n.
寄生虫；此处指疟原虫

infected adj.
受感染的

indicate v.
表明；说明

voluntarily adv.
自愿地

thereafter adv.
此后

patient n.
患者

obtain v.
获得

crystal n.
晶体

recipient n.
获奖者

jury n.
评审团

physiology n.
生理学

laureate n.
获奖者；此处指诺奖得主

guideline n.
指南

therapy n.
疗法

deadly adj.
最致命的

skip v.
略过；省去

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 This _____ boiling.

02 Efficacy in mice indicated nothing about human _____.

03 In July 1972, to assess its _____, Tu and two colleagues voluntarily took the extract.

04 The _____ of Ge Hong's line lay in the boiling it skipped.

文章主线

- 任务 523项目联合攻关，目标是对旧药氯喹有抗药性的疟疾；屠呦呦带队查2000多药方，试约380种提取物。
- 线索 青蒿药效不稳；葛洪古方浸泡青蒿而不煮沸，启发屠呦呦怀疑高温会破坏有效成分。
- 突破 改用沸点约35°C的乙醚低温提取，1971年第191号样品完全抑制小鼠疟原虫；人体安全仍待验证。
- 验证 1972年，她与两名同事试服提取物，再用于21名疟疾患者，随后得到青蒿素晶体。
- 影响 以青蒿素为基础的多药合用疗法获世卫推荐，挽救众多生命；屠呦呦2015年获诺奖。

参考答案　1 bypassed 2 safety 3 toxicity 4 essence